Saipranith Swargam

Backend Engineer | Distributed Systems, Kafka & Elasticsearch | Python, Go, Kubernetes

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Summary

Backend engineer (∼2 years incl. internship) who owns services end to end, from design to production metrics and alerts, in **Python** and **Go** on Kafka, Elasticsearch, PostgreSQL and Kubernetes. Made a key API 11x faster.

Experience

Software Engineer I, Cyware Aug 2025 – Present

- Built a Python ingestion microservice that polls external data feeds and publishes normalized batches to **Kafka**; a streaming packer cut messages from 3,844 to 604, bytes 850 to 361 MB and CPU 21.5 to 8.9 s.
- Cut a heavy read API from 4,240 to 385 ms on a 10.5M-document dataset by replacing **Elasticsearch** scrolls with composite aggregations and EXPLAIN-tuned **PostgreSQL** queries (lookup 1,074 to 36 ms).
- Built a resumable, checkpointed data-seeding pipeline that loads ∼155k records per tenant as a Kubernetes post-deploy job; regression tests stored all 62,911 relationships, 0 of 1,542 batches failed.
- Instrumented two event-driven **Go** services with OpenTelemetry and Prometheus metrics, health probes and alert rules across 6 production clusters; fixed a misfiring alert to zero false alerts.
- Scaled a third-party polling pipeline to tenants with 60+ monitored assets by splitting one combined fetch into 4 parallel jobs with configurable batching, keeping each run within serverless time limits.
- Designed a read-time ID mapping in a **Go** event-history service so history survives record merges across Postgres and S3 archives, restoring history for 721 of 888 merged records in testing without a backfill.
- Refactored a **Python** integrations service of 60+ third-party connectors around a shared REST-request helper, migrating ∼53 connectors and fixing resource leaks, with phased QA plans per step.
- Reworked an agentic AI workflow to cut LLM token usage by 90% while preserving response quality.
- Fixed AWS SES rate limiting with a queue-based email sending system, reducing send failures by 40%.

Software Engineering Intern, Cyware Dec 2024 – Jul 2025

- Built **Prometheus** failure metrics for asynchronous Celery jobs with a standalone multiprocess exporter, rolled out across 5 Helm charts, catching failures early and reducing customer-reported problems by 30%.
- Cut API response latency by 83% through query and API tuning, including a **PostgreSQL** GIN trigram index for text search and splitting a heavy aggregate endpoint into 4 lean, index-friendly views.
- Built multi-method login (SAML, LDAP and OTP) end to end across the **Django** auth proxy and per-tenant auth config, with client-specific error codes and a migration script to retire a legacy flag.

Full Stack Developer, Bitezy *(food ordering platform, remote)* Mar 2024 – Aug 2024

- Built WebSocket live order tracking for restaurants, customers and delivery partners, improving communication efficiency by 40%, and tuned the system to handle 1,000+ live orders (INR 4 lakh revenue).

Projects

SwiftFrame: near-real-time stream processing Jan 2024 – Feb 2024

- Built a stream processing engine on **Kafka** and **Flink** with PostgreSQL sinks and Grafana monitoring, reaching 955 MB/s throughput at 130 ms latency on 60K records/min; first-author paper at ICCCNT 2024.

Technical Skills

Languages: Python, Go, TypeScript, JavaScript, SQL, Java, C++, Dart
Backend & Data: Django, REST APIs, Kafka, NATS JetStream, Elasticsearch, PostgreSQL, ClickHouse, Redis
Infra & Observability: Kubernetes, Docker, Helm, Argo CD, GitHub Actions, AWS, OpenTelemetry, Prometheus
Other: Flink, Next.js, React, Node.js, WebSockets, Celery, Claude Code

Education

B.Tech, Computer Science and Engineering, IIIT Sri City (CGPA **8.74**/10) Dec 2021 – May 2025

Honors & Awards

- **Above & Beyond Award**, Cyware (2025). **Winner**, Cyware hackathon (2025) and Flipr Hackathon 18 (2023).